

# MANEESH KUMAR SINGH

Computational Researcher · Scientific ML & Data Assimilation · Numerical PDE & FEM · Data-Driven Scientific Computing

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## RESEARCH INTERESTS

- Numerical Methods for PDEs • Stochastic & Multiscale Modelling • Data Assimilation • Probabilistic Machine Learning

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## TECHNICAL SKILLS & OPEN-SOURCE

- **Languages:** Python (primary), MATLAB, C++
- **ML / Scientific Libraries:** NumPy, SciPy, PyTorch, JAX, ParaView
- **FEM & PDE Solvers:** Firedrake, FEniCS, deal.II, ParMooN
- **HPC & Parallel Computing:** PETSc, MPI
- **Dev Tools:** Git, GitHub Actions, Linux/Unix, L<sup>A</sup>T<sub>E</sub>X, Jupyter
- **GitHub:** <https://github.com/maneeshkrsingh>
- **Personal Research Site:** <https://maneeshkrsingh.github.io>

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## EMPLOYMENT

- SRM Institute of Science and Technology, Chennai • Research Assistant Professor • since 2026  
*Focus: Bayesian data assimilation, Particle filtering, Score-based Generative Modelling*
- Imperial College London • Postdoctoral Research Associate • 2022–2025  
Mentors: Colin Cotter, Dan Crisan, Darryl Holm  
*Focus: Bayesian data assimilation, Stochastic PDE, Particle filtering, compatible FEM for GFD*
- University of Konstanz • Postdoctoral Fellow • 2021–2022  
Mentor: Stefan Frei  
*Focus: Space-time FEM for PDEs on moving domains*
- Indian Institute of Science, Bangalore • Postdoctoral Fellow • 2019–2021  
Mentor: Sashikumaar Ganesan  
*Focus: Operator-splitting FEM, Radiative-transfer equations, ParMooN parallel solver*

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## EDUCATION

- Indian Institute of Technology Guwahati • PhD in Mathematics • 2013–2019  
Thesis: *Analysis of robust computational techniques for singularly perturbed system of parabolic partial differential equations*  
Supervisor: Prof. Natesan Srinivasan
- University of Hyderabad • MSc in Applied Mathematics • 2011–2013
- Banaras Hindu University • BA (Hons) in Mathematics • 2008–2011

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#### FELLOWSHIPS & AWARDS

- D.S. Kothari Postdoctoral Fellowship • UGC, India • 2021
- NBHM Postdoctoral Fellowship • DAE, India • 2018
- CSIR-UGC JRF and GATE Fellowship in Mathematics • MHRD, India • 2013

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#### WORKING PAPERS

- M. K. Singh, C. J. Cotter, D. Crisan: Data assimilation for a stochastic Euler equation using nudged particle filter.
- D. D. Holm, M. K. Singh, C. J. Cotter: Energy conserving compatible FEM discretisations for deterministic and stochastic rotating shallow water equations.
- M. K. Singh, O. D. Street: Energy-consistent numerical integrator of wave equation with time-dependent bathymetry.
- S. Frei, M. K. Singh: Eulerian time-stepping method for transient transport problems on moving domains.

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#### SUBMITTED ARTICLES

- M. K. Singh, J. Hope-Collins, C. J. Cotter, D. Crisan: Data assimilation using a global Girsanov nudged particle filter.  
Submitted to *Statistics and Computing* • <https://arxiv.org/abs/2507.17685>

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#### PUBLISHED ARTICLES

13. D. D. Holm, M. K. Singh, O. D. Street: A comparative numerical study of stochastic Hamiltonian Camassa-Holm equations.  
*Proceedings of the Royal Society A* • 2026
12. D. D. Holm, M. K. Singh: A multiscale Camassa-Holm equation.  
*Chaos, Solitons and Fractals* • 2026
11. C. J. Cotter, D. Crisan, M. K. Singh: Data assimilation for the stochastic Camassa-Holm equation using particle filtering: a numerical investigation.  
*Stochastic Transport in Upper Ocean Dynamics III, Springer Nature* • 2024
10. S. Frei, M. K. Singh: An implicitly extended Crank-Nicolson scheme for the heat equation on a time-dependent domain.  
*Journal of Scientific Computing* • 2024
9. S. Ganesan, M. K. Singh: Operator-splitting finite element method for solving the radiative transfer equation.  
*Numerical Algorithms* • 2024
8. M. K. Singh: A discrete-ordinate weak Galerkin method for radiative transfer equation.  
*Applied Numerical Mathematics* • 2024
7. M. K. Singh, G. Singh, S. Natesan: A unified study on superconvergence analysis of Galerkin FEM for singularly perturbed systems of multiscale nature.  
*Journal of Applied Mathematics and Computing* • 2021

6. S. Natesan, M. K. Singh: Robust computational method for singularly perturbed system of parabolic convection-diffusion problems with interior layers.  
*Computational and Mathematical Methods* • 2020
5. M. K. Singh, S. Natesan: Numerical solution of 2D singularly perturbed reaction-diffusion system with multiple scales.  
*Computers and Mathematics with Applications* • 2020
4. M. K. Singh, S. Natesan: A parameter-uniform hybrid finite difference scheme for singularly perturbed system of parabolic convection-diffusion problems.  
*International Journal of Computer Mathematics* • 2019
3. M. K. Singh, S. Natesan: Numerical analysis of singularly perturbed system of parabolic convection-diffusion problem with regular boundary layers.  
*Differential Equations and Dynamical Systems* • 2019
2. M. K. Singh, S. Natesan: A robust computational method for singularly perturbed system of 2D parabolic convection-diffusion problems.  
*International Journal of Mathematical Modelling and Numerical Optimisation* • 2019
1. M. K. Singh, S. Natesan: Richardson extrapolation technique for singularly perturbed system of parabolic PDEs with exponential boundary layers.  
*Applied Mathematics and Computation* • 2018

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#### CONFERENCE PROCEEDINGS

- C. J. Cotter, D. Crisan, M. K. Singh: The GONG Particle Filter: A Novel Framework for Data Assimilation in Stochastic PDEs.  
*CASML 2024, IISc Bangalore*
- M. K. Singh, S. Natesan: A finite element superconvergence approximation for singularly perturbed system of convection-diffusion problems.  
*AIP Conference Proceedings, FIAM 2018*

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#### TEACHING

- SRMIST, Chennai • Course Instructor • Probability and Queuing Theory • 2026–
- Imperial College London • Substitute Instructor • Computational Linear Algebra, Finite Elements • 2022–2025
- IIT Guwahati • Teaching Fellow • Calculus, Linear Algebra & ODE, Scientific Computing, PDE, Computational Finance • 2014–2018

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#### CONFERENCE TALKS

- LACAM-2025, IISER Thiruvananthapuram • 2025
- MATHSD-2025, IIITM Gwalior • 2025
- PEML2025, London • 2025
- 6th STUOD Annual Workshop, University of Edinburgh • 2025
- Firedrake Meeting 2025, University of Leeds • 2025
- ENUMATH 2025, Heidelberg • 2025

- 5th STUOD Annual Workshop, INRIA Rennes • 2024
  - Firedrake Meeting 2024, University of Oxford • 2024
  - 4th STUOD Annual Workshop, IFREMER, France • 2023
  - Firedrake Meeting 2023, London • 2023
  - ADENA, IIT Guwahati • 2021
  - FIAM 2018, NIT Hamirpur • 2018
  - 62nd ISTAM Congress, Hyderabad • 2017
  - ICMMDSCA-2016, IIT Kanpur • 2016
  - ICTIMCAMS, BHU Varanasi • 2016
  - ICCTP:TC, SAU New Delhi • 2015
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#### INVITED TALKS

- FoCuS-ML, IIITM Gwalior • 2025
  - TIFR-CAM Seminar • 2024
  - IIIT Gwalior Workshop on PDE • 2024
  - ICL/UCL Numerics Seminar • 2023
  - 24th STUOD Sandbox Workshop, Imperial College London • 2023
  - Research Scholar Seminar, IIT Guwahati • 2022
  - Oberseminar Numerik, University of Konstanz • 2021
  - Workshop on Parallel FE Computing using ParMooN, IISc Bangalore • 2019
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#### REFERENCES

- Prof. Natesan Srinivasan (PhD Supervisor) • IIT Guwahati • [natesan@iitg.ac.in](mailto:natesan@iitg.ac.in)
- Prof. Colin Cotter (Postdoc Mentor) • Imperial College London • [colin.cotter@imperial.ac.uk](mailto:colin.cotter@imperial.ac.uk)
- Prof. Darryl D. Holm (Postdoc Mentor) • Imperial College London • [d.holm@imperial.ac.uk](mailto:d.holm@imperial.ac.uk)